Canonical Bundle Formula

We work over an algebraically closed field of characteristic zero, like $\mathbb{C}$. All varieties are assumed to be normal unless otherwise stated.

1 Birational divisors

Definition 1 (b-divisors)**.** Let X be a normal algebraic variety. We call Y a *birational model over* X, or simply a *model over* X, if there exists a proper birational morphism $Y \to X$ of normal varieties. When both X and Y are quasi-projective, $Y \to X$ is projective.

Let $X' \dashrightarrow X$ be a birational map. For any prime divisor E (resp. algebraic valuation v) over X, we denote by $c_{X'}(E)$ (resp. $v_{X'}$) the centre of E (resp. v) on X', whenever it exists. A *b-divisor* $\mathbf{D}$ on X is a formal sum

$$\mathbf{D} = \sum_E r_E E \qquad \left(\text{resp. } \sum_v r_v v\right),$$

where the E are prime divisors over X (resp. the v are algebraic valuations over X) and $r_E \in \mathbb{R}$ (resp. $r_v \in \mathbb{R}$), such that only finitely many terms with nonzero coefficient have a divisorial centre on X. If, in addition, $r_E \in \mathbb{Q}$ for every E (resp. $r_v \in \mathbb{Q}$ for every v), then $\mathbf{D}$ is called a $\mathbb{Q}$-*b-divisor* on X.

The *trace* of $\mathbf{D}$ on a model $X' \to X$ is the $\mathbb{R}$-divisor

$$\operatorname{tr}_{X'} \mathbf{D} = D_{X'} = \sum_{E' \text{ a divisor on } X'} r_{E'} E' \qquad \left(\text{resp. } \sum_{v_{X'} \text{ a divisor}} r_v v_{X'}\right).$$

If $D_{X'}$ is $\mathbb{R}$-Cartier and D_Y is the pullback of $D_{X'}$ on Y for every model $Y \to X'$, we say that $\mathbf{D}$ *descends to* X' and write

$$\mathbf{D} = \overline{D_{X'}}.$$

Let $X \to U$ be a proper morphism, and assume that $\mathbf{D}$ is a b-divisor over X which descends to some model $Y \to X$. If D_Y is nef$/U$ (resp. semi-ample$/U$), then we say that $\mathbf{D}$ is *b-nef*$/U$ (resp. *b-semi-ample*$/U$). If D_Y is an $\mathbb{R}$-Cartier divisor, then we say that $\mathbf{D}$ is $\mathbb{R}$-*b-Cartier*. The terms b-Cartier and $\mathbb{Q}$-b-Cartier are defined similarly using Cartier and $\mathbb{Q}$-Cartier divisors, respectively. If $\mathbf{D}$ can be written as an $\mathbb{R}_{\geq 0}$-linear combination of nef$/U$ $\mathbb{Q}$-b-Cartier b-divisors, then we say that $\mathbf{D}$ is NQC/U.

When $U = \operatorname{Spec} k$, this gives the absolute setting.

Definition 2 (Generalized pairs)**.** A *generalized sub-pair* (or *g-sub-pair* for short) is a triple $(X, B, \mathbf{M})/U$ consisting of a normal algebraic variety X associated with a proper morphism $X \to U$, an $\mathbb{R}$-divisor B on X, and an $\mathbb{R}$-b-Cartier b-nef$/U$ divisor $\mathbf{M}$ on X, such that $K_X + B + M_X$ is $\mathbb{R}$-Cartier. Note that, on a model $h : W \to X$ on which $\mathbf{M}$ descends, we put

$$M_X := h_* M_W.$$

If B is a $\mathbb{Q}$-divisor and $\mathbf{M}$ is $\mathbb{Q}$-b-Cartier, then we say that $(X, B, \mathbf{M})/U$ is a $\mathbb{Q}$-*g-sub-pair*.

If $\mathbf{M} = 0 = \overline{0}$, a g-sub-pair $(X, B, \mathbf{M})/U$ is called a *sub-pair*. If $U = \{\text{pt}\}$, or if U is not important, we drop U from the notation.

A g-sub-pair (resp. $\mathbb{Q}$-g-sub-pair) $(X, B, \mathbf{M})/U$ is called a *g-pair* (resp. $\mathbb{Q}$-*g-pair*) if $B \geq 0$. A sub-pair (X, B) is called a *pair* if $B \geq 0$.

Definition 3 (Singularities of g-pairs)**.** Let $(X, B, \mathbf{M})/U$ be a g-sub-pair. For any prime divisor E on X and any $\mathbb{R}$-divisor D on X, we define $\operatorname{mult}_E D$ to be the multiplicity of D along E.

Let $h : W \to X$ be any log resolution of (X, B) such that $\mathbf{M}$ descends to W, and write

$$K_W + B_W + M_W = h^*(K_X + B + M_X).$$

The *log discrepancy* of a prime divisor E on W with respect to $(X, B, \mathbf{M})$ is $1 - \operatorname{mult}_E B_W$, and it is denoted by $A(E, X, B, \mathbf{M})$. For the discrepancy

$$a(E, X, B, \mathbf{M}) = -\operatorname{mult}_E B_W,$$

we have

$$A(E, X, B, \mathbf{M}) = 1 + a(E, X, B, \mathbf{M}).$$

We say that $(X, B, \mathbf{M})$ is *sub-g-lc* (resp. *sub-g-klt*) if

$$A(E, X, B, \mathbf{M}) \geq 0 \qquad (\text{resp. } > 0)$$

for every log resolution $h : W \to X$ as above and every prime divisor E on W.

Suppose that $(X, B, \mathbf{M})$ is sub-g-lc. A *g-lc place* of $(X, B, \mathbf{M})$ is a prime divisor E over X such that

$$A(E, X, B, \mathbf{M}) = 0.$$

A *g-lc centre* of $(X, B, \mathbf{M})$ is the centre of a g-lc place on X. We often say *lc place* and *lc centre* for short when the pair is clear.

We say that $(X, B, \mathbf{M})$ is *(sub-)g-dlt* if it is (sub-)g-lc and there exists an open subset $U \subseteq X$ such that

(1) $(U, B|_U)$ is log smooth;

(2) for any lc place E over X, we have $c_X(E) \cap U \neq \varnothing$, and $c_X(E) \cap U$ is an lc centre of $(U, B|_U)$.

There is an important result in [Bir-Zhang14, proof of Lemma 4.5], also proved in [HL22, Proposition 3.9].

Definition–Lemma 4 (G-dlt modifications)**.** Let $(X, B, \mathbf{M})/U$ be a quasi-projective g-lc g-pair. Then there is a projective birational morphism $r : X' \to X$ and a g-lc g-pair $(X', B', \mathbf{M})/U$ such that

(1) $(X', B', \mathbf{M})$ is $\mathbb{Q}$-factorial g-dlt;

(2) $K_{X'} + B' + M_{X'} = r^*(K_X + B + M_X)$; and

(3) any r-exceptional divisor is a component of $\lfloor B' \rfloor$.

For any projective birational morphism r and $(X', B', \mathbf{M})/U$ which satisfy (1)–(3), $(r, (X', B', \mathbf{M})/U)$ will be called a *g-dlt modification* of $(X, B, \mathbf{M})$.

It has a useful application; see [JLX22, Lemma 2.6] or my note [Adj, 9, 10, 14, 15].

Definition–Lemma 5 (G-dlt adjunction)**.** Let $(X, B, \mathbf{M})/U$ be a g-lc g-pair and let $S \subseteq \lfloor B \rfloor$ be a prime divisor, with normalization $\nu : S^\nu \to S$. Then there is a naturally defined g-pair structure $(S^\nu, B_{S^\nu}, \mathbf{M}^S)/U$ such that

$$K_{S^\nu} + B_{S^\nu} + M^S_{S^\nu} \sim_{\mathbb{R}} (K_X + B + M_X)|_{S^\nu}.$$

We say that $(S^\nu, B_{S^\nu}, \mathbf{M}^S)/U$ is the g-pair obtained by *divisorial adjunction*. If $(X, B, \mathbf{M})$ is g-lc, then $(S^\nu, B_{S^\nu}, \mathbf{M}^S)$ is g-lc, and if $(X, B, \mathbf{M})$ is g-dlt, then $(S^\nu, B_{S^\nu}, \mathbf{M}^S)$ is g-dlt.

For the construction of divisorial adjunction, see [Adj, 9 and 10]; for the g-dlt and g-lc parts, see [Adj, 14 and 15].

Definition 6 (Discrepancy b-divisors)**.** Let $(X, B, \mathbf{M})/U$ be a g-sub-pair. For each model $f : Y \to X$, choose compatible canonical divisors with $f_*K_Y = K_X$, and set

$$\operatorname{tr}_Y \mathbf{A}(X, B, \mathbf{M}) = K_Y + M_Y - f^*(K_X + B + M_X).$$

These traces define the *discrepancy b-divisor* $\mathbf{A}(X, B, \mathbf{M})$; we often write its trace A_Y for short. Define another b-divisor by

$$\operatorname{tr}_Y \mathbf{A}^*(X, B, \mathbf{M}) = A_Y^{>-1},$$

where we omit the components with coefficient at most -1. For a prime divisor E over X,

$$\operatorname{mult}_E \mathbf{A}(X, B, \mathbf{M}) = A(E, X, B, \mathbf{M}) - 1 = a(E, X, B, \mathbf{M}),$$

where $a(E, X, B, \mathbf{M})$ is the discrepancy of E.

Definition 7 (Sheaf associated to a b-divisor)**.** Let $\mathbf{D}$ be an $\mathbb{R}$-b-divisor on X. The sheaf $\mathcal{O}_X(\mathbf{D})$ is an $\mathcal{O}_X$-module whose sections on an open subset $U \subseteq X$ are given by

$$H^0\big(U, \mathcal{O}_X(\mathbf{D})\big) = \big\{\varphi \in k(X)^* \;\big|\; \operatorname{div}_Y(\varphi) + D_Y \geq 0 \text{ on } h^{-1}(U) \text{ for every model } h : Y \to X\big\} \cup \{0\},$$

where $k(X)$ is the function field of X. Note that $\mathcal{O}_X(\mathbf{D})$ is not necessarily coherent.

Lemma 8. *When* $\mathbf{D}$ *is b-Cartier,* $\mathcal{O}_X(\mathbf{D})$ *is coherent.*

Proof. Take a model $h : W \to X$ on which $\mathbf{D}$ descends. Then we have

$$\mathcal{O}_X(\mathbf{D}) = h_*\mathcal{O}_W(D_W).$$

Indeed, the inclusion $\mathcal{O}_X(\mathbf{D}) \subseteq h_*\mathcal{O}_W(D_W)$ is trivial. For the other inclusion, suppose that $\varphi \in H^0(h^{-1}(V), \mathcal{O}_W(D_W))$. This means that

$$\operatorname{div}_W(\varphi) + D_W \geq 0 \qquad \text{on } h^{-1}(V).$$

For any other model $Y \to X$, take a high common model

$$W \xleftarrow{\ r\ } T \xrightarrow{\ s\ } Y$$

over X. Since $\mathbf{D}$ descends to W,

$$\operatorname{div}_T(\varphi) + D_T = r^*\big(\operatorname{div}_W(\varphi) + D_W\big) \geq 0$$

on $(h \circ r)^{-1}(V)$. Pushing forward via s gives

$$\operatorname{div}_Y(\varphi) + D_Y = s_*\big(\operatorname{div}_T(\varphi) + D_T\big) \geq 0$$

on $s((h \circ r)^{-1}(V))$. Thus $h_*\mathcal{O}_W(D_W) \subseteq \mathcal{O}_X(\mathbf{D})$, and coherence follows from the properness of h. □

For a b-divisor $\mathbf{D}$ on X, its *round up* $\lceil \mathbf{D} \rceil$ is defined by

$$\mathrm{tr}_Y \lceil \mathbf{D} \rceil = \lceil D_Y \rceil$$

on every model $Y \to X$. Similarly for round down.

Definition 9 (Contraction)**.** A *contraction* is a proper surjective morphism $f : X \to Y$ of normal varieties with $f_* \mathcal{O}_X = \mathcal{O}_Y$.

Equivalently, f has connected fibres. Indeed, in the Stein factorization

$$X \longrightarrow \widetilde{Y} \longrightarrow Y,$$

connected fibres force the finite morphism $\widetilde{Y} \to Y$ to have degree one, i.e., it is a birational morphism between normal varieties, so it is an isomorphism.

Definition 10 (G-lc-trivial fibration)**.** Let $(X, B, \mathbf{M})/U$ be a g-sub-pair, and let $f : X \to Y$ be a contraction over U. We call $f : (X, B, \mathbf{M}) \to Y$ a *g-lc-trivial fibration* over U if

(1) $(X, B, \mathbf{M})$ is g-sub-lc over the generic point of Y;

(2) $\operatorname{rank} f_* \mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil\big) = 1$; and

(3) there exists an $\mathbb{R}$-Cartier divisor L on Y such that

$$K_X + B + M_X \sim_{\mathbb{R}} f^* L.$$

If, in addition, $(X, B, \mathbf{M})$ is g-sub-klt over the generic point of Y, we say *g-klt-trivial*. For $\mathbf{M} = 0$, these are classical lc-trivial and klt-trivial fibrations; see [FG14, Definitions 3.1–3.2].

Remark 11. Condition (3) is sometimes written as

$$K_X + B + M_X \sim_{\mathbb{R}, Y} 0.$$

When f is a contraction, they are equivalent.

Lemma 12 ([Fuj12, Lemmas 3.19–3.20])**.** *Let $(X, B, \mathbf{M})/U$ be a g-sub-lc g-sub-pair, and let $h : W \to X$ be a log resolution on which $\mathbf{M}$ descends. Write*

$$K_W + B_W + M_W = h^*(K_X + B + M_X).$$

Then

$$\mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil\big) = h_* \mathcal{O}_W\big(\lceil -B_W^{<1} \rceil\big).$$

In particular, this sheaf is coherent. If $B \geq 0$, then

$$\mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil\big) = \mathcal{O}_X.$$

Proof. By definition,

$$\mathrm{tr}_W \, \mathbf{A}(X, B, \mathbf{M}) = -B_W, \qquad \mathrm{tr}_W \, \mathbf{A}^*(X, B, \mathbf{M}) = -B_W^{<1}.$$

Let $g : W' \to W$ be any model over W. Define $B_{W'}$ by

$$K_{W'} + B_{W'} + M_{W'} = (h \circ g)^*(K_X + B + M_X).$$

Since $\mathbf{M}$ descends to W, we have $M_{W'} = g^* M_W$. Cancelling these terms gives

$$K_{W'} + B_{W'} = g^*(K_W + B_W).$$

Step 1. We claim that

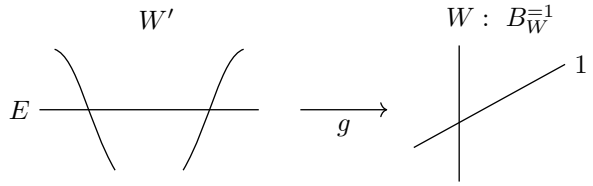
$$\lceil \mathbf{A}^*(X, B, \mathbf{M})_{W'} \rceil = g^* \lceil \mathbf{A}^*(X, B, \mathbf{M})_W \rceil + F,$$

where $F \geq 0$ is an integral g-exceptional divisor.

The divisor $\lceil -B_W^{<1} \rceil$ is an effective Cartier divisor, and is supported on the negative components of B_W. Thus the pair

$$\big(W, B_W + \lceil -B_W^{<1} \rceil\big)$$

is log smooth and has lc singularities, since for a coefficient $b < 1$, we have $0 \leq b + \lceil -b \rceil < 1$. It also has the same lc places as (W, B_W), because an lc place of either log smooth pair has centre a stratum of $B_W^{=1}$, but $\operatorname{Supp} \lceil -B_W^{<1} \rceil$ contains no such stratum.



$$a(E) = 2 - 1 - (1 + 1) = -1$$

For a prime divisor E on W', the number $\operatorname{mult}_E g^* \lceil -B_W^{<1} \rceil$ is a nonnegative integer, since $g^* \lceil -B_W^{<1} \rceil \geq 0$. If $\operatorname{mult}_E B_{W'} = 1$, then E is an lc place of (W, B_W), so its multiplicity in $g^* \lceil -B_W^{<1} \rceil$ is zero. Both sides of the claimed equality

$$\lceil -B_{W'}^{<1} \rceil = g^* \lceil -B_W^{<1} \rceil + F$$

then give $\operatorname{mult}_E F = 0$.

If $\operatorname{mult}_E B_{W'} < 1$, we have

$$A(E, W, B_W) = 1 - \operatorname{mult}_E B_{W'} > 0,$$

so E is not an lc place of (W, B_W). Since the two pairs have the same lc places, we have

$$\begin{aligned} 0 &< A\big(E, W, B_W + \lceil -B_W^{<1} \rceil\big) \\ &= 1 - \operatorname{mult}_E B_{W'} - \operatorname{mult}_E g^* \lceil -B_W^{<1} \rceil. \end{aligned}$$

Rounding this inequality gives

$$\lceil -\operatorname{mult}_E B_{W'} \rceil \geq \operatorname{mult}_E g^* \lceil -B_W^{<1} \rceil \geq 0.$$

The left-hand side is nothing but $\operatorname{mult}_E \lceil -B_{W'}^{<1} \rceil$, so $\operatorname{mult}_E F \geq 0$. Since (W, B_W) is sub-lc, the coefficients of $B_{W'}$ are at most 1. Therefore $F \geq 0$ is effective. Since both sides of the claimed equality are integral divisors, F is integral.

Finally, if E is not g-exceptional, then $g(E)$ is a divisor on W. That is, g is an isomorphism over the generic point of $g(E)$ by normality, so we have

$$\operatorname{mult}_E B_{W'} = \operatorname{mult}_{g(E)} B_W.$$

Then

$$\operatorname{mult}_E F = \operatorname{mult}_E \lceil -B_{W'}^{<1} \rceil - \operatorname{mult}_E g^* \lceil -B_W^{<1} \rceil = 0.$$

We see that F is g-exceptional. This proves the claim.

Step 2. By algebraic Hartogs, we have

$$g_* \mathcal{O}_{W'}(F) = \mathcal{O}_W.$$

Then we have

$$\begin{aligned} g_* \mathcal{O}_{W'}\big(\lceil \mathbf{A}^*(X, B, \mathbf{M})_{W'} \rceil\big) &= g_* \mathcal{O}_{W'}\big(g^* \lceil -B_W^{<1} \rceil + F\big) \\ &\cong \mathcal{O}_W\big(\lceil -B_W^{<1} \rceil\big), \end{aligned}$$

thanks to the projection formula. These identifications are induced by the identity on rational functions.

For any other model $\psi : Y \to X$, take a common higher model

$$W \xleftarrow{r} T \xrightarrow{s} Y.$$

Fix any open subset $V \subseteq X$. Suppose that we have

$$\operatorname{div}_W(\varphi) + \lceil -B_W^{<1} \rceil \geq 0$$

on $h^{-1}(V)$. Applying the claimed equality to $r : T \to W$, we have

$$r_* \mathcal{O}_T\big(\lceil \mathbf{A}^*(X, B, \mathbf{M})_T \rceil\big) \cong \mathcal{O}_W\big(\lceil -B_W^{<1} \rceil\big).$$

Thus it gives

$$\operatorname{div}_T(\varphi) + \lceil \mathbf{A}^*(X, B, \mathbf{M})_T \rceil \geq 0$$

on $(h \circ r)^{-1}(V)$. We push forward this along $s : T \to Y$ and get

$$\operatorname{div}_Y(\varphi) + \lceil \mathbf{A}^*(X, B, \mathbf{M})_Y \rceil = s_*\big(\operatorname{div}_T(\varphi) + \lceil \mathbf{A}^*(X, B, \mathbf{M})_T \rceil\big) \geq 0$$

on $s((h \circ r)^{-1}(V)) = \psi^{-1}(V)$. It gives

$$h_* \mathcal{O}_W\big(\lceil -B_W^{<1} \rceil\big) \subseteq \mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil\big).$$

The reverse direction is trivial. Coherence is due to the properness of h.

Finally, if $B \geq 0$, then every non-exceptional component of B_W has coefficient in $[0, 1]$. Hence $\lceil -B_W^{<1} \rceil$ is effective and h-exceptional. By algebraic Hartogs, we have

$$h_* \mathcal{O}_W\big(\lceil -B_W^{<1} \rceil\big) = \mathcal{O}_X.$$

□

Now let $f : X \to Y$ be a contraction over U. If $(X, B, \mathbf{M})$ is only g-sub-lc over the generic point of Y, i.e., it is g-sub-lc over an open subset $Y^0 \subseteq Y$, applying Lemma 12, we have

$$\mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil\big) = h_* \mathcal{O}_W\big(\lceil -B_W^{<1} \rceil\big)$$

on $f^{-1}(Y^0)$. Pushing forward along f, we get

$$f_* \mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil\big) = (f \circ h)_* \mathcal{O}_W\big(\lceil -B_W^{<1} \rceil\big)$$

on Y^0, and since $\eta_Y \in Y^0$, we have

$$\operatorname{rank} f_* \mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil\big) = \operatorname{rank}(f \circ h)_* \mathcal{O}_W\big(\lceil -B_W^{<1} \rceil\big).$$

Lemma 13 (The rank-one condition)**.** *Let $(X, B, \mathbf{M})/U$ be a g-sub-pair, and let $f : X \to Y$ be a contraction over U. Suppose that $(X, B, \mathbf{M})$ is g-sub-lc over the generic point of Y and that B is effective over that point, i.e., its components which dominate Y have nonnegative coefficients. Then*

$$\operatorname{rank} f_*\mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M})\rceil\big) = 1.$$

Proof. We may replace Y by an open subset and assume that $B \geq 0$ and $(X, B, \mathbf{M})$ is g-lc; this does not change the rank. Take $h : W \to X$ to be a log resolution on which $\mathbf{M}$ descends. The divisor $\lceil -B_W^{<1}\rceil$ is effective and h-exceptional. So by Lemma 12,

$$\operatorname{rank} f_*\mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M})\rceil\big) = \operatorname{rank} f_*\mathcal{O}_X = \operatorname{rank} \mathcal{O}_Y = 1.$$

□

Example 14 ($B \not\geq 0$ over the generic point)**.** Put $f : \mathbb{P}^1 \to \operatorname{Spec} \mathbb{C}$, $B = -p$, and $\mathbf{M} = 0$, where $p \in \mathbb{P}^1$. The pair $(\mathbb{P}^1, B)$ is a sub-klt pair. Take $h = \mathrm{id}$. Then

$$h_*\mathcal{O}_{\mathbb{P}^1}\big(\lceil -B^{<1}\rceil\big) = \mathcal{O}_{\mathbb{P}^1}(p).$$

Therefore,

$$\operatorname{rank} f_*\mathcal{O}_{\mathbb{P}^1}(p) = \dim_{\mathbb{C}} H^0\big(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(p)\big) = 2,$$

not rank one.

2 Divisorial Thresholds: Discriminant and Moduli b-divisor

For a contraction $f : (X, B, \mathbf{M}) \to Y$ where $(X, B, \mathbf{M})$ is g-sub-lc over the generic point of Y, we fix an $\mathbb{R}$-Cartier divisor L on Y and an $\mathbb{R}$-linear equivalence

$$K_X + B + M_X \sim_{\mathbb{R}} f^*L.$$

Let P be a prime divisor over Y. Choose a smooth model $\psi : Y' \to Y$ on which P appears. Resolve the main component of $X \times_Y Y'$ and take a sufficiently high model W on which $\mathbf{M}$ descends. We obtain the commutative diagram

$$\begin{array}{ccc} W & \xrightarrow{h} & X \\ {\scriptstyle f_W}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[\psi]{} & Y. \end{array}$$

We put

$$K_W + B_W + M_W = h^*(K_X + B + M_X).$$

Definition 15. The *generalized lc threshold* over η_P, the generic point of P, is

$$t_P = \sup\big\{s \in \mathbb{R} \mid (W, B_W + sf_W^*P, \mathbf{M}) \text{ is g-sub-lc over } \eta_P\big\}.$$

The condition is tested over an open neighbourhood of η_P in Y', and the parameter s may be negative.

Lemma 16 (Threshold formula)**.** *After replacing W by a higher log resolution, we may assume that* $\operatorname{Supp} B_W \cup \operatorname{Supp} f_W^*P$ *is snc over η_P. Let $E_1, \dots, E_n$ be the components of this union meeting the fibre $f_W^{-1}(\eta_P)$, and put*

$$b_i = \operatorname{mult}_{E_i} B_W, \qquad m_i = \operatorname{mult}_{E_i} f_W^*P.$$

Then

$$t_P = \min_{m_i>0} \frac{1-b_i}{m_i}. \tag{2.1}$$

The value of t_P is independent of the choice of log resolution.

Proof. Since $\mathbf{M}$ descends to W, the generalized discrepancies coincide with the discrepancies of $(W, B_W + sf_W^*P)$. The snc criterion gives that this sub-pair is sub-lc over η_P if and only if $b_i + sm_i \le 1$ for every i. Since f_W is proper and E_i meets the fibre over η_P, its image is either P or Y'. If $m_i = 0$, the component E_i dominates Y', and $b_i \le 1$ by generic sub-lc. If $m_i > 0$, the condition is $s \le (1-b_i)/m_i$. At least one such component exists because f_W is surjective and P is an effective Cartier divisor. This proves (2.1).

On a common higher resolution over Y', the crepant pullbacks of $K_W + B_W + sf_W^*P + M_W$ compute the same discrepancies. Hence t_P is independent of the resolution. See [BGR, Lemma 5.2.16] for the discrepancy comparison under crepant pullback for pairs. □

Definition 17 (Discriminant and moduli part)**.** For each prime divisor P over Y, put $\operatorname{mult}_P \mathbf{C} = 1 - t_P$. On a smooth model $\psi : Y' \to Y$, define

$$C_{Y'} = \sum_{P \subset Y'} (1-t_P)P, \qquad N_{Y'} = \psi^*L - K_{Y'} - C_{Y'},$$

where the sum runs over the prime divisors on Y'. The b-divisor $\mathbf{C}$ is called the *discriminant*, and $\mathbf{N} = \overline{L} - \mathbf{K} - \mathbf{C}$ is the *moduli b-divisor*, where $\mathbf{K}$ has traces given by compatible canonical divisors on the models of Y.

On other models their traces are obtained by pushforward from a dominating smooth model.

Proposition 18. $\mathbf{C}$ *and* $\mathbf{N}$ *are b-divisors, and on every model $\psi : Y' \to Y$ their traces satisfy*

$$K_{Y'} + C_{Y'} + N_{Y'} = \psi^*L.$$

Proof. Step 1. Fix a smooth model $\psi : Y' \to Y$. Choose a log resolution $h : W \to X$ mapping to Y' on which $\mathbf{M}$ descends and $\operatorname{Supp} B_W$ is snc. We use the same diagram

$$\begin{array}{ccc} W & \xrightarrow{h} & X \\ {\scriptstyle f_W}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[\psi]{} & Y. \end{array}$$

By generic smoothness, there is a nonempty open subset $U \subseteq Y'$ such that $f_W^{-1}(U) \to U$ is smooth. By properness of f_W, we may shrink U further to cut off the vertical components of B_W. After shrinking U further, the remaining horizontal components have relative snc support over U. Their intersections are smooth over U, and their restrictions to every (possibly non-closed) fibre have snc support.

For a prime divisor P on Y' meeting U, its generic point η_P lies in U. Take an open subset $\eta_P \in V \subseteq U$ such that $P_V = P \cap V$ is smooth, and set $W_V = f_W^{-1}(V)$. The pullback divisor over V has the following scheme-theoretic description.[1]

$$D := f_W^* P|_{W_V} \cong (P \times_{Y'} W) \times_W (V \times_{Y'} W) \cong W_V \times_V P_V.$$

Since $W_V \to V$ is smooth, $D \to P_V$ is smooth. As P_V is smooth, D is smooth and hence reduced. In particular, $f_W^* P$ is reduced over η_P.

At any point of D, let $H_1, \dots, H_r$ be the components of $\operatorname{Supp} B_W$ passing through that point, and set $S = \bigcap_{i=1}^r H_i$ (with $S = W$ if $r = 0$). Write $S_V = S \cap W_V$. We have

$$S \cap D \cong S \times_W (W_V \times_V P_V) \cong S \times_{Y'} P_V \cong S_V \times_V P_V.$$

This is the scheme-theoretic inverse image of P_V under $f_W|_{S_V} : S_V \to V$. Since $S \cap f_W^{-1}(U) \to U$ is smooth by the relative snc condition, $S_V \to V$ is smooth. Thus $S \cap D$ is smooth and, where nonempty, is an effective Cartier divisor in S_V, by smooth base change of the smooth Cartier divisor P_V. Therefore D meets the horizontal boundary strata transversely.

On W_V write

$$B_W|_{W_V} = \sum_i b_i H_i, \qquad D = \sum_j D_j,$$

where D is reduced and the H_i are the horizontal components restricted to W_V. Each D_j maps into P_V, so no D_j is a component of $B_W|_{W_V}$. Then

$$B_W|_{W_V} + sD = \sum_i b_i H_i + \sum_j s D_j,$$

where every $b_i \le 1$ because $(X, B, \mathbf{M})$ is g-sub-lc over the generic point of Y. Since $\mathbf{M}$ descends to W, the g-sub-pair $(W, B_W + s f_W^* P, \mathbf{M})$ is g-sub-lc over η_P if and only if the sub-pair $(W, B_W + s f_W^* P)$ is sub-lc over η_P. This sub-pair is log smooth over η_P by the transversality proved above. From the displayed coefficients, it is sub-lc over η_P if and only if $s \le 1$. Thus $t_P = 1$. Only the finitely many prime divisors in $Y' \setminus U$ can contribute to $C_{Y'}$.

Step 2. Let $r : Y'' \to Y'$ be a higher smooth model. If $\widetilde{P}$ is the strict transform of P, they define the same divisorial valuation. Since r is an isomorphism over a neighbourhood of η_P, the thresholds agree: $t_{\widetilde{P}} = t_P$. The r-exceptional terms push forward to zero, so

$$r_* C_{Y''} = \sum_{P \subset Y'} (1 - t_{\widetilde{P}}) r_* \widetilde{P} = \sum_{P \subset Y'} (1 - t_P) P = C_{Y'}.$$

Choose compatible canonical divisors, i.e., traces of $\mathbf{K}$. Then

$$r_* N_{Y''} = r_* \big(r^* \psi^* L - K_{Y''} - C_{Y''} \big) = \psi^* L - K_{Y'} - C_{Y'} = N_{Y'}.$$

A common higher smooth model therefore makes the pushforwards to any normal model independent of the smooth model used to compute them. This proves that $\mathbf{C}$ and $\mathbf{N}$ are b-divisors. Pushing forward the identity on a smooth model also gives the stated identity on every model. □

[1] Here we use the scheme-theoretic notation $P_V = P \cap V \cong P \times_{Y'} V$ and $W_V = f_W^{-1}(V) \cong V \times_{Y'} W$. The displayed identification follows by transitivity of fibre products [Görtz I, Proposition 4.16].

Note that if

$$L' = L + \sum_j r_j \operatorname{div}_Y(u_j), \qquad r_j \in \mathbb{R}, \quad u_j \in k(Y)^\times,$$

then the same discriminant $\mathbf{C}$ occurs for L' and

$$N'_{Y'} = N_{Y'} + \sum_j r_j \operatorname{div}_{Y'}(u_j)$$

on each model $\psi : Y' \to Y$, where the u_j are viewed as rational functions on Y'. Thus changing L by an $\mathbb{R}$-principal divisor changes $\mathbf{N}$ by the corresponding $\mathbb{R}$-principal b-divisor.

Theorem 19 (Canonical bundle formula)**.** *Let $f : X \to Y$ be a contraction of normal quasi-projective varieties over U. Let $(X, B, \mathbf{M})/U$ be a g-pair which is g-lc over the generic point of Y, and suppose that*

$$K_X + B + M_X \sim_{\mathbb{R}} f^*L,$$

where L is an $\mathbb{R}$-Cartier divisor on Y. Then the moduli b-divisor $\mathbf{N}$ is b-nef/U and $\mathbb{R}$-b-Cartier.

Proof. By Lemma 13, $f : (X, B, \mathbf{M}) \to Y$ is a g-lc-trivial fibration. Applying [CHLX23, Theorem 11.4.4] gives the result. □

Hashizume gives an alternative proof in [Has26, Theorem 1.3]. This theorem requires no NQC condition.

Proposition 20. *Under the hypotheses of Theorem 19, we have $C_Y \geq 0$ and $(Y, C_Y, \mathbf{N})/U$ is a g-pair satisfying*

$$K_Y + C_Y + N_Y = L,$$

where $N_Y = \operatorname{tr}_Y \mathbf{N}$. For every prime divisor P over Y,

$$A(P, Y, C_Y, \mathbf{N}) = t_P.$$

Moreover, if $(X, B, \mathbf{M})$ is a g-lc (resp. g-klt) g-pair, then $(Y, C_Y, \mathbf{N})$ is a g-lc (resp. g-klt) g-pair.

Proof. Step 1. First we prove that $C_Y \geq 0$. Let P be a prime divisor on Y. Since Y is normal, we may choose an open subset $V \subseteq Y$ containing η_P on which $P|_V$ is Cartier. After shrinking V, write

$$P|_V = \operatorname{div}_V(\varphi)$$

as an effective Cartier divisor. Put $V' = f^{-1}(V)$ and $f_V = f|_{V'} : V' \to V$. Its pullback is

$$f_V^*(P|_V) = \operatorname{div}_{V'}(\varphi \circ f).$$

Since f is proper and surjective, this divisor has a prime component dominating $P|_V$. Let $E \subseteq X$ be its closure in X, and put

$$m = \operatorname{mult}_E f_V^*(P|_V) > 0, \qquad b = \operatorname{mult}_E B \geq 0.$$

Adding $sf_V^*(P|_V)$ to $B|_{V'}$ gives multiplicity $b+sm$ along E. Thus g-sub-lc over η_P requires $b + sm \le 1$. We may compute the threshold on a sufficiently high model on which $\mathbf{M}$ descends and the crepant pullback has snc support over η_P. Consequently,

$$t_P \le \frac{1-b}{m} \le 1, \qquad \operatorname{mult}_P C_Y = 1 - t_P \ge 0,$$

where m is a positive integer. This holds for every prime divisor P on Y, so $C_Y \ge 0$.

Step 2. By definition, $N_Y = L - K_Y - C_Y$. The divisor L is $\mathbb{R}$-Cartier, and Theorem 19 gives that $\mathbf{N}$ is b-nef/U and $\mathbb{R}$-b-Cartier. Hence $(Y, C_Y, \mathbf{N})/U$ is a g-pair.

Let P be any prime divisor over Y. Choose a smooth model $\psi : Y' \to Y$ on which P appears and $\mathbf{N}$ descends. The definition of $N_{Y'}$ gives

$$K_{Y'} + C_{Y'} + N_{Y'} = \psi^* L = \psi^*(K_Y + C_Y + N_Y).$$

Thus $C_{Y'}$ is the crepant boundary used to compute the log discrepancies of $(Y, C_Y, \mathbf{N})$. Since $\operatorname{mult}_P C_{Y'} = 1 - t_P$, we have

$$A(P, Y, C_Y, \mathbf{N}) = 1 - \operatorname{mult}_P C_{Y'} = t_P.$$

Step 3. It remains to prove the singularity assertions. Suppose that $(X, B, \mathbf{M})$ is g-lc. For the prime divisor $P \subseteq Y'$ chosen in Step 2, take a log resolution $h : W \to X$ with a commutative diagram

$$\begin{array}{ccc} W & \xrightarrow{g} & Y' \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle \psi} \\ X & \xrightarrow[f]{} & Y \end{array}$$

on which $\mathbf{M}$ descends and $\operatorname{Supp} B_W \cup \operatorname{Supp} g^*P$ has snc support. Let $E_1, \dots, E_n$ be the components of g^*P dominating P, and set

$$m_i = \operatorname{mult}_{E_i} g^*P > 0, \qquad b_i = \operatorname{mult}_{E_i} B_W \le 1.$$

Since $\mathbf{M}$ descends to W, the g-sub-pair $(W, B_W + sg^*P, \mathbf{M})$ is g-sub-lc over η_P if and only if the sub-pair $(W, B_W + sg^*P)$ is sub-lc over η_P. By the snc condition, this is equivalent to $b_i + sm_i \le 1$ for every i; see [BGR, Corollary 5.2.20]. Lemma 16 gives

$$t_P = \min_{m_i > 0} \frac{1 - b_i}{m_i} \ge 0.$$

Thus $(Y, C_Y, \mathbf{N})$ is g-lc. If $(X, B, \mathbf{M})$ is g-klt, every $b_i < 1$, so the same formula gives $t_P > 0$. Hence $(Y, C_Y, \mathbf{N})$ is g-klt. □

Example 21. Let E be an elliptic curve, let Y be a smooth projective variety, and let $P \subseteq Y$ be a smooth prime divisor. Set $f : E \times Y \to Y$, $B = bf^*P$, and $\mathbf{M} = 0$, where $b \ge 0$. Choose a nowhere-vanishing differential form on E, so its canonical divisor is zero. Then

$$K_{E\times Y} + B \sim_{\mathbb{R}} f^*(K_Y + bP).$$

The pair $(E \times Y, B)$ is klt over the generic point of Y because the boundary is vertical. Since $f^*P = E \times P$ is smooth and reduced, and

$$B + sf^*P = (b + s)(E \times P),$$

Lemma 16 gives $t_P = 1 - b$.

For every prime divisor $Q \subseteq Y$ different from P, the boundary B has empty support over η_Q, so $t_Q = 1$. Taking $L = K_Y + bP$, we obtain

$$C_Y = bP, \qquad N_Y = L - K_Y - C_Y = 0.$$

For example, if $b = 2$, the pair $(Y, 2P)$ is not lc, although $(E \times Y, 2f^*P)$ is klt over the generic point of Y, and $t_P = -1$.

Lemma 22 (Restriction of a divisorial valuation)**.** *Let $f : X \to Y$ be a dominant morphism of algebraic varieties. Let E be a prime divisor over X, and suppose that $\operatorname{ord}_E|_{k(Y)}$ is nontrivial. Then there are a model $\psi : Y' \to Y$, a prime divisor $P \subseteq Y'$, and an integer $e > 0$ such that*

$$\operatorname{ord}_E|_{k(Y)} = e \operatorname{ord}_P .$$

Here the restriction is the valuation

$$v : k(Y)^\times \longrightarrow \mathbb{Z}, \qquad u \longmapsto \operatorname{ord}_E\big((f \circ h)^* u\big),$$

where $h : W \to X$ is a model carrying E.

Proof. Consider the valuation rings

$$R_E = \mathcal{O}_{W,\eta_E}, \qquad R_v = R_E \cap k(Y),$$

where we identify $k(Y) \subseteq k(X) = k(W)$ using the dominant morphism f and the birational morphism h. Denote their residue fields by

$$k(E) = R_E/\mathfrak{m}_E, \qquad k(v) = R_v/\mathfrak{m}_v.$$

The value group of v is a nonzero subgroup of $\mathbb{Z}$, so R_v is a discrete valuation ring.

Step 1. We first prove

$$s := \operatorname{trdeg}_{k(v)} k(E) \le \operatorname{trdeg}_{k(Y)} k(X) = \dim X - \dim Y.$$

Lift algebraically independent elements $\overline{u}_1, \dots, \overline{u}_s$ of $k(E)$ over $k(v)$ to $u_1, \dots, u_s \in R_E$. These lifts have value zero, since they lie outside $\mathfrak{m}_E$, and are algebraically independent over $k(Y)$. This proves the inequality.

Since $E \subseteq W$ is a prime divisor, we have

$$\begin{aligned} d := \operatorname{trdeg}_k k(v) &= \operatorname{trdeg}_k k(E) - \operatorname{trdeg}_{k(v)} k(E) \\ &= \dim X - 1 - s \ge \dim Y - 1. \end{aligned}$$

For the reverse inequality, lift a transcendence basis $\overline{z}_1, \dots, \overline{z}_d$ of $k(v)/k$ to $z_1, \dots, z_d \in R_v$. Each lift has value zero. Choose $t \in k(Y)$ with $v(t) > 0$. For a polynomial

$$\sum_i a_i t^i, \qquad a_i \in k[z_1, \dots, z_d] \setminus \{0\},$$

we have

$$v(a_i t^i) = v(a_i) + i\, v(t) = i\, v(t).$$

Let i_0 be the least exponent appearing in the sum. We write

$$\sum_i a_i t^i = t^{i_0}\left(a_{i_0} + \sum_{i>i_0} a_i t^{i-i_0}\right),$$

where

$$\overline{a_{i_0} + \sum_{i>i_0} a_i t^{i-i_0}} = \overline{a}_{i_0} \neq 0 \qquad \text{in } k(v).$$

Thus the term in parentheses is nonzero, which implies that $\sum_i a_i t^i \neq 0$. Hence $z_1, \dots, z_d, t$ are algebraically independent over k, i.e., $d+1 \leq \dim Y$. So

$$\operatorname{trdeg}_k k(v) = \dim Y - 1.$$

Step 2. Let $x \in X$ be the centre of R_E on X, i.e., the generic point of $c_X(E)$, and put $y = f(x)$. The inclusion

$$\mathcal{O}_{Y,y} \longrightarrow R_E \cap k(Y) = R_v$$

is a local homomorphism, so R_v has centre y on Y. By [BGR, Lemma 5.2.13], a finite sequence of blow-ups of its successive centres gives a proper birational morphism $Y_n \to Y$ and a point $y_n \in Y_n$ such that

$$\mathcal{O}_{Y_n,y_n} \cong R_v.$$

The corresponding morphisms fit in the diagram

$$\begin{array}{ccccccccc} Y & \leftarrow & Y_1 & \longleftarrow & Y_2 & \longleftarrow & \cdots & \leftarrow & Y_n \ni y_n \\ & \nwarrow & & & & & & \nearrow & \\ & & & & \operatorname{Spec} R_v & & & & \end{array}$$

where $\mathfrak{m}_v \in \operatorname{Spec} R_v$ maps to the successive centres, and

$$\operatorname{trdeg}_k k(y_n) = \operatorname{trdeg}_k k(v) = \dim Y - 1.$$

Step 3. Take the normalization $Y' \to Y_n$. Since R_v is a DVR and hence normal, the local ring at the point above y_n is still R_v. This is a codimension-one point, whose closure is a prime divisor $P \subseteq Y'$ satisfying

$$\mathcal{O}_{Y',\eta_P} \cong \mathcal{O}_{Y_n,y_n} \cong R_v.$$

Let e be the value under v of a uniformizer of $\mathcal{O}_{Y',\eta_P}$. Then $e > 0$ and $v = e \operatorname{ord}_P$, as required. □

Proposition 23 (Images of lc centres)**.** *Assume that* $(X, B, \mathbf{M})$ *in Theorem 19 is a g-lc g-pair. Then the lc centres of* $(Y, C_Y, \mathbf{N})$ *are exactly the images of the vertical lc centres of* $(X, B, \mathbf{M})$*.*

Proof. Let P be an lc place of $(Y, C_Y, \mathbf{N})$. By Proposition 20, $t_P = 0$. Lemma 16 gives that this minimum is attained by a component E with

$$\operatorname{mult}_E f_W^* P > 0, \qquad \operatorname{mult}_E B_W = 1$$

on a log resolution $h : W \to X$ mapping to a model Y' carrying P, on which $\mathbf{M}$ descends. We have the diagram

$$\begin{array}{ccc} W & \xrightarrow{f_W} & Y' \supseteq P \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle \psi} \\ X & \xrightarrow[f]{} & Y. \end{array}$$

Then E is an lc place of $(X, B, \mathbf{M})$ and dominates P. Its centre $c_X(E)$ maps onto $c_Y(P)$.

Conversely, let E be an lc place of $(X, B, \mathbf{M})$ whose centre on X does not dominate Y. Put

$$Z = f(c_X(E)) \subsetneq Y.$$

Let $x \in X$ be the generic point of $c_X(E)$ and $y = f(x)$. Choose a nonzero $u \in \mathfrak{m}_{Y,y}$, viewed as a rational function regular near y and vanishing along Z. The local homomorphisms

$$\mathcal{O}_{Y,y} \longrightarrow \mathcal{O}_{X,x} \longrightarrow R_E$$

send u into $\mathfrak{m}_E$, so $\operatorname{ord}_E((f \circ h)^* u) > 0$. Thus $\operatorname{ord}_E|_{k(Y)}$ is nontrivial. By Lemma 22, it is $e \operatorname{ord}_P$ for a prime divisor P over Y.

Take a smooth model $\psi : Y' \to Y$ carrying P, and a log resolution $h : W \to X$ carrying E, on which $\mathbf{M}$ descends, with a morphism $f_W : W \to Y'$. Identify $k(Y) = k(Y')$ and put $y' = f_W(\eta_E)$. There is a local homomorphism

$$\mathcal{O}_{Y',y'} \longrightarrow R_E \cap k(Y') = R_v,$$

so y' is a centre of v. Since $R_v \cong \mathcal{O}_{Y',\eta_P}$, the point η_P is also its centre. As Y' is separated, the centre is unique. Therefore $f_W(E) = P$. Moreover, if u is a local equation of P near η_P, with $\operatorname{ord}_P(u) = 1$, then

$$\begin{aligned} \operatorname{mult}_E f_W^* P &= \operatorname{ord}_E(f_W^* u) = v(u) \\ &= e \operatorname{ord}_P(u) = e > 0. \end{aligned}$$

Since E is an lc place, $A(E, X, B, \mathbf{M}) = 0$, and hence

$$\operatorname{mult}_E(B_W + s f_W^* P) = 1 + se.$$

Thus g-sub-lc over η_P requires $1 + se \le 1$, i.e., $s \le 0$. The assumption that $(X, B, \mathbf{M})$ is g-lc permits $s = 0$, so

$$A(P, Y, C_Y, \mathbf{N}) = t_P = 0.$$

The centre of P on Y is $f(c_X(E))$, which proves the claim. □

Lemma 24. *Let $\pi : W \to U$ be a morphism of quasi-projective varieties, with W smooth, and let A_W be a semi-ample/U $\mathbb{R}$-Cartier divisor. Fix a projective birational morphism $r : V \to W$ with V smooth, and an snc divisor Θ on V with irreducible components $E_1, \dots, E_s$. Then there is an effective $\mathbb{R}$-divisor $H \sim_{\mathbb{R},U} A_W$ such that*

$$0 \le \operatorname{mult}_Z r^* H < 1 \qquad \text{for every prime divisor } Z \subseteq V,$$

while, in particular,

$$\operatorname{mult}_{E_i} r^* H = 0 \qquad (i = 1, \dots, s),$$

and $\Theta + r^* H$ has snc support.

Proof. Since A_W is semi-ample/U, there is a morphism $g : W \to Z$ over U such that $A_W \sim_{\mathbb{R}} g^*A$, where A is an ample/U $\mathbb{R}$-Cartier divisor on Z. Let $\pi_Z : Z \to U$ be the structure morphism. Choose an ample $\mathbb{R}$-Cartier divisor B on U such that $A + \pi_Z^*B$ is ample. We may write

$$A + \pi_Z^*B = \sum_j c_j A_j,$$

where $c_j \in \mathbb{R}_{>0}$ and the A_j are ample Cartier divisors. Choose integers $n_j > c_j$ such that $n_j A_j$ is very ample. Set

$$L_j = g^*(n_j A_j), \qquad c'_j = \frac{c_j}{n_j} \in (0,1), \qquad A_W + \pi^*B \sim_{\mathbb{R}} \sum_j c'_j L_j.$$

Choose finite-dimensional subspaces

$$U_j \subseteq H^0(Z, \mathcal{O}_Z(n_j A_j)), \qquad U_j \otimes_k \mathcal{O}_Z \twoheadrightarrow \mathcal{O}_Z(n_j A_j).$$

The pullbacks of their generators generate $\mathcal{O}_W(L_j)$ and $\mathcal{O}_V(r^*L_j)$. Choose general members $D_j \in |U_j|$, viewed as divisors on W, such that $\Theta + \sum_j r^*D_j$ has snc support and

$$\operatorname{mult}_{E_i} r^*D_j = 0 \qquad \text{for all } i, j.$$

The divisors r^*D_j are reduced and have no common components. Then

$$H = \sum_j c'_j D_j$$

is as desired, since

$$H \sim_{\mathbb{R}} A_W + \pi^*B, \qquad \text{hence} \qquad H \sim_{\mathbb{R},U} A_W,$$

and

$$\operatorname{mult}_Z r^*H \in \{0\} \cup \{c'_j\}_j.$$

□

Lemma 25. *Let $f_W : W \to Y'$ be a surjective projective morphism over a quasi-projective variety U, with W smooth and quasi-projective, on which $\mathbf{M}$ descends. Let $P \subseteq Y'$ be an irreducible Cartier divisor, and let D_W be an $\mathbb{R}$-divisor such that (W, D_W) is sub-klt over the generic point of Y'. Put*

$$c = t_P = \sup\{s \in \mathbb{R} \mid (W, D_W + sf_W^*P) \text{ is sub-lc over } \eta_P\}.$$

If A_W is a semi-ample/U $\mathbb{R}$-Cartier divisor, there is an effective $H \sim_{\mathbb{R},U} A_W$ such that

$$c = \sup\{s \in \mathbb{R} \mid (W, D_W + H + sf_W^*P) \text{ is sub-lc over } \eta_P\},$$

and $(W, D_W + H)$ is still sub-klt over the generic point of Y'.

Proof. Take a log resolution $r : V \to W$ of $(W, D_W + f_W^*P)$, and define D_V by

$$K_V + D_V = r^*(K_W + D_W).$$

Lemma 16 says that $D_V + c(f_W \circ r)^* P$ has coefficients at most one over η_P, and there is a component Z dominating P with

$$\operatorname{mult}_Z\big(D_V + c(f_W \circ r)^* P\big) = 1.$$

Put $\Theta = D_V + c(f_W \circ r)^* P$. Apply the previous lemma to $r : V \to W$ and the reduced divisor supported on $\operatorname{Supp} D_V \cup \operatorname{Supp}(f_W \circ r)^* P$. We obtain an effective $H \sim_{\mathbb{R},U} A_W$ such that $\Theta + r^* H$ has snc support, every coefficient of $r^* H$ is less than one, and every component of $\operatorname{Supp} D_V \cup \operatorname{Supp}(f_W \circ r)^* P$ has multiplicity zero in $r^* H$. Therefore the snc condition gives that

$$\big(V, D_V + r^* H + c(f_W \circ r)^* P\big) \text{ is sub-lc over } \eta_P.$$

Moreover, $\operatorname{mult}_Z r^* H = 0$ and $\operatorname{mult}_Z (f_W \circ r)^* P > 0$, so for $s > c$ we have

$$\operatorname{mult}_Z\big(D_V + r^* H + s(f_W \circ r)^* P\big) = 1 + (s - c) \operatorname{mult}_Z (f_W \circ r)^* P > 1.$$

Thus the new threshold with respect to $D_V + r^* H$ is still c. It is easy to see that $(W, D_W + H)$ is sub-klt over the generic point of Y'. □

3 Abundance

Let D be an $\mathbb{R}$-Cartier divisor on a normal quasi-projective variety X over U. For a prime divisor P over X, choose a model $r : V \to X$ on which P appears. The *asymptotic vanishing order* of D along P is

$$o_P(D) = \inf\big\{\operatorname{mult}_P G \mid G \in |r^* D/U|_{\mathbb{R}}\big\},$$

where

$$|r^* D/U|_{\mathbb{R}} = \big\{G \geq 0 \mid G \sim_{\mathbb{R},U} r^* D \text{ on } V\big\}, \qquad \inf \varnothing = +\infty.$$

Pullback and pushforward on a common higher model show that this definition is independent of the chosen model.

Assume that $X \to U$ is projective. Recall that the *invariant Iitaka dimension* of D over U is

$$\kappa_\iota(X/U, D) = \begin{cases} \kappa(X/U, Z), & D \sim_{\mathbb{R},U} Z \text{ for some } Z \geq 0, \\ -\infty, & \text{if no such } Z \text{ exists.} \end{cases}$$

Here the existence of Z is tested after restricting to a nonempty open subset of the image of X in U. Equivalently,

$$\kappa_\iota(X/U, D) = \kappa_\iota(X_{\bar\eta}, D|_{X_{\bar\eta}}),$$

where $X_{\bar\eta}$ is the geometric generic fibre of the contraction in the Stein factorization of $X \to U$.

If D is nef/U, put $n = \dim X_{\bar\eta}$ and take an ample/U Cartier divisor H on X. The *numerical dimension* of D over U is

$$\begin{aligned} \nu(X/U, D) &= \nu(X_{\bar\eta}, D|_{X_{\bar\eta}}) \\ &= \max\big\{0 \leq k \leq n \mid (D|_{X_{\bar\eta}})^k \cdot (H|_{X_{\bar\eta}})^{n-k} > 0\big\}. \end{aligned}$$

Take a smooth model $r : V \to X$. For a Cartier divisor A on X, put

$$\begin{aligned} \sigma(D/U; A) &= \sigma(D|_{X_{\bar\eta}}; A|_{X_{\bar\eta}}) \\ &= \max\left\{k \in \mathbb{Z}_{\geq 0} \;\middle|\; \limsup_{m \to \infty} \frac{h^0\big(V_{\bar\eta}, \mathcal{O}_{V_{\bar\eta}}((\lfloor m r^* D \rfloor + r^* A)|_{V_{\bar\eta}})\big)}{m^k} > 0\right\}, \end{aligned}$$

An empty set has value $-\infty$. The *sigma dimension* is

$$\kappa_\sigma(X/U, D) = \max_A \sigma(D/U; A),$$

where A ranges over Cartier divisors on X. In particular, if D is nef/U, then $\kappa_\sigma(X/U, D) = \nu(X/U, D)$.

An $\mathbb{R}$-Cartier divisor D is *abundant over* U if $\kappa_\iota(X/U, D) = \kappa_\sigma(X/U, D)$. If D is nef/U, this is equivalent to $\kappa_\iota(X/U, D) = \nu(X/U, D)$. See [Hu20, Section 2.3] and [b-Div].

For a nef/U $\mathbb{R}$-Cartier divisor D, the following conditions are equivalent.

(1) D is abundant over U.

(2) There are a model $p : X' \to X$, a surjective morphism $f : X' \to Y$ of smooth quasi-projective varieties over U, and a nef and big/U $\mathbb{R}$-Cartier divisor B on Y such that $p^*D \sim_{\mathbb{R}} f^*B$.

(3) For a general fibre F of the contraction in the Stein factorization of $X \to U$, we have $o_P(D|_F) = 0$ for every prime divisor P over F.

For the proof, see [Hu20, Lemma 2.8].

Lemma 26. *Let A and A' be $\mathbb{R}$-Cartier divisors on a normal quasi-projective variety X over U, with $A = A' + E$, where $E \geq 0$. Suppose $|A'/U|_{\mathbb{R}} \neq \varnothing$. For a prime divisor P on X,*

$$0 \leq o_P(A) \leq o_P(A') + \operatorname{mult}_P E.$$

The same assertion holds for a prime divisor P over X, using the pullback of E to a model on which P appears.

Proof. Take $G \geq 0$ with $A' \sim_{\mathbb{R},U} G$. Then $G + E$ is effective and $G + E \sim_{\mathbb{R},U} A$, so $o_P(A) \leq \operatorname{mult}_P G + \operatorname{mult}_P E$. Taking the infimum over G proves the assertion. □

Definition 27 (Ambro model)**.** Let $f : (X, B) \to Y$ be an lc-trivial fibration over U, and fix an $\mathbb{R}$-Cartier divisor L on Y defining its moduli b-divisor $\mathbf{N}$. A model $\psi : Y' \to Y$ is an *Ambro model* for f if $N_{Y'}$ is nef/U and $\mathbb{R}$-Cartier and, for every higher model $r : Y'' \to Y'$, we have

$$N_{Y''} = r^* N_{Y'}.$$

In other words, $\mathbf{N}$ descends to Y' with nef/U trace.

A criterion for an Ambro model. Let $f : (X, B) \to Y$ be an lc-trivial fibration of normal quasi-projective varieties over U. Fix a convex decomposition

$$B = \sum_i \beta_i \Gamma_i, \qquad \beta_i > 0, \quad \sum_i \beta_i = 1,$$

where each Γ_i is a $\mathbb{Q}$-divisor and each $f : (X, \Gamma_i) \to Y$ is a $\mathbb{Q}$-lc-trivial fibration. Consider a commutative diagram

$$\begin{array}{ccc} W & \xrightarrow{h} & X \\ {\scriptstyle f_W}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[\psi]{} & Y, \end{array}$$

where W and Y' are smooth models and f_W is a contraction over U. For each i, define $\Gamma_{i,W}$ by

$$K_W + \Gamma_{i,W} = h^*(K_X + \Gamma_i).$$

We require the following conditions.

(1) For each i, $f_W : (W, \Gamma_{i,W}) \to Y'$ is a $\mathbb{Q}$-lc-trivial fibration, and there are $\phi_i \in k(W)^\times$ and a $\mathbb{Q}$-Cartier divisor H_i on Y' with

$$K_W + \Gamma_{i,W} + \frac{1}{r_i} \operatorname{div}_W(\phi_i) = f_W^* H_i,$$

where r_i is the least positive integer for which $r_i(K_{\overline{F}} + \Gamma_{i,W}|_{\overline{F}}) \sim 0$ on the geometric generic fibre $\overline{F}$ of f_W.

(2) There are reduced snc divisors S on W and T on Y', so (W, S) and (Y', T) are log smooth. The vertical part of S is the reduced divisor supported on $f_W^{-1}(\operatorname{Supp} T)$.

(3) Put $V = Y' \setminus \operatorname{Supp} T$. The pair $(f_W^{-1}(V), S|_{f_W^{-1}(V)})$ is log smooth over V, i.e., for every stratum Θ of this pair, the restriction $f_W|_\Theta : \Theta \to V$ is smooth.

(4) The divisors $\Gamma_{i,W}$ and $\operatorname{div}_W(\phi_i)$ are supported on S. The traces on Y' of the discriminant and moduli b-divisors $\mathbf{C}_i$ and $\mathbf{N}_i$ of $f_W : (W, \Gamma_{i,W}) \to Y'$, with moduli trace defined using H_i, are supported on T.

In [Hu20, Definition 3.7], this is called a resolved model of $f_W : (W, B_W) \to Y'$, where $B_W = \sum_i \beta_i \Gamma_{i,W}$. By [Hu20, Remark 3.9], Y' is an Ambro model for f_W.

The following proposition gives a rational decomposition compatible with the discriminant and moduli b-divisors. See [Hu20, Proposition 3.6].

Proposition 28 (Rational decomposition of an lc-trivial fibration)**.** *Let $f : (X, B) \to Y$ be an lc-trivial fibration over U of normal quasi-projective varieties. Fix an $\mathbb{R}$-Cartier divisor L on Y with $K_X + B \sim_{\mathbb{R}} f^*L$, and let $\mathbf{C}, \mathbf{N}$ be the discriminant and moduli b-divisors defined using L.*

Then there are $\mathbb{Q}$-divisors B_i on X, $\mathbb{Q}$-Cartier divisors L_i on Y, and real numbers $a_i > 0$ with $\sum_i a_i = 1$ such that

$$B = \sum_i a_i B_i, \qquad L = \sum_i a_i L_i, \qquad K_X + B_i \sim_{\mathbb{Q}} f^* L_i,$$

where each $f : (X, B_i) \to Y$ is a $\mathbb{Q}$-lc-trivial fibration. If $\mathbf{C}_i, \mathbf{N}_i$ are its discriminant and moduli b-divisors defined using L_i, then

$$\mathbf{C} = \sum_i a_i \mathbf{C}_i, \qquad \mathbf{N} = \sum_i a_i \mathbf{N}_i.$$

If the geometric generic fibre pair $(X_{\bar\eta}, B|_{X_{\bar\eta}})$ of f is klt, the pairs $(X_{\bar\eta}, B_i|_{X_{\bar\eta}})$ can also be chosen klt.

Proof. Step 1. We set up the parameters for B_i, L_i. Write $L = \sum_k \ell_k A_k$ with A_k Cartier on Y. Choose $\varphi_j \in k(X)^\times$ and $u_j \in \mathbb{R}$ such that

$$K_X + B + \sum_j u_j \operatorname{div}_X(\varphi_j) = \sum_k \ell_k f^* A_k.$$

For arbitrary real numbers y_k, v_j, put $L_\Gamma = \sum_k y_k A_k$ and define

$$\Gamma = f^*L_\Gamma - K_X - \sum_j v_j \operatorname{div}_X(\varphi_j).$$

The values $y_k = \ell_k$ and $v_j = u_j$ give (B, L). If all y_k, v_j are rational, then Γ, L_Γ are $\mathbb{Q}$-divisors and $K_X + \Gamma \sim_{\mathbb{Q}} f^*L_\Gamma$.

Step 2. Take a log resolution $h : W \to X$ of the union of the supports of B, the f^*A_k and the $\operatorname{div}_X(\varphi_j)$. The difference $\Gamma - B$ is supported on these additional divisors, so h also resolves every (X, Γ) in Step 1. Define B_W and Γ_W by

$$K_W + B_W = h^*(K_X + B), \qquad K_W + \Gamma_W = h^*(K_X + \Gamma).$$

For each horizontal component E of the snc divisor on W, including the horizontal exceptional divisors, set

$$b_E = \operatorname{mult}_E B_W, \qquad c_E = \operatorname{mult}_E \Gamma_W, \qquad d_E = \max\{0, \lceil -b_E \rceil\}.$$

Let y_k, v_j vary subject to $-d_E \le c_E \le 1$ for every such E. Then (X, Γ) is sub-lc over the generic point η of Y.

On the generic fibre W_η of $f \circ h$, set

$$D_\Gamma = \lceil -\Gamma_W^{<1} \rceil|_{W_\eta}, \qquad D_B = \lceil -B_W^{<1} \rceil|_{W_\eta}.$$

The coefficient of $\lceil -\Gamma_W^{<1} \rceil$ along E is $\max\{0, \lceil -c_E \rceil\}$, which lies in $[0, d_E]$. Thus $0 \le D_\Gamma \le D_B$. Taking global sections gives

$$k(Y) \subseteq H^0(W_\eta, \mathcal{O}_{W_\eta}(D_\Gamma)) \subseteq H^0(W_\eta, \mathcal{O}_{W_\eta}(D_B)). \tag{3.1}$$

By Lemma 12, over a neighbourhood of η,

$$\mathcal{O}_X(\lceil \mathbf{A}^*(X, B) \rceil) = h_* \mathcal{O}_W(\lceil -B_W^{<1} \rceil).$$

Taking the generic stalk after pushing forward by f, the rank-one condition for $f : (X, B) \to Y$ gives $h^0(W_\eta, \mathcal{O}_{W_\eta}(D_B)) = 1$. Applying the same sheaf formula to Γ and using (3.1), we get

$$\operatorname{rank} f_* \mathcal{O}_X(\lceil \mathbf{A}^*(X, \Gamma) \rceil) = 1.$$

Hence $f : (X, \Gamma) \to Y$ is lc-trivial, and is $\mathbb{Q}$-lc-trivial when all the parameters are rational.

Denote $p = ((y_k)_k, (v_j)_j)$ and $p_0 = ((\ell_k)_k, (u_j)_j)$. Take a large enough positive integer M and set

$$\mathcal{P} = \left\{ p \;\middle|\; \begin{array}{l} -d_E \le c_E \le 1 \text{ for every horizontal component } E, \\ |y_k| \le M, \quad |v_j| \le M \text{ for every } k, j \end{array} \right\}.$$

If $(X_{\bar\eta}, B|_{X_{\bar\eta}})$ is klt, each $b_E < 1$. Choose rational numbers q_E with $b_E < q_E < 1$. Impose the additional conditions

$$\mathcal{P}_{\mathrm{klt}} = \left\{ p \in \mathcal{P} \;\middle|\; c_E \le q_E \text{ for every such } E, \quad \Gamma|_{X_\eta} \ge 0 \right\}.$$

The last condition consists of nonnegativity inequalities on the coefficients of the horizontal components of Γ on X. These are rational polytopes containing p_0 in their respective cases. Choose a convex decomposition in $\mathcal{P}$ or $\mathcal{P}_{\mathrm{klt}}$ as appropriate,

$$p_0 = \sum_{i=1}^{v} a_i p_i, \qquad a_i > 0, \qquad \sum_{i=1}^{v} a_i = 1,$$

where the p_i are rational points. Each p_i gives (Γ_i, L_i) defining a $\mathbb{Q}$-lc-trivial fibration $f : (X, \Gamma_i) \to Y$, with

$$K_X + \Gamma_i \sim_{\mathbb{Q}} f^* L_i, \qquad B = \sum_i a_i \Gamma_i, \qquad L = \sum_i a_i L_i.$$

Step 3. We construct W, Y', S, T as in the resolved model above, with W dominating the log resolution in Step 2. For each i, we write $K_W + \Gamma_{i,W} = h^*(K_X + \Gamma_i)$. Since $K_X + \Gamma_i \sim_{\mathbb{Q}} f^* L_i$, there is a least positive integer r_i such that

$$r_i(K_{\overline{F}} + \Gamma_i|_{\overline{F}}) \sim 0$$

on the geometric generic fibre $\overline{F}$ of f. The index is the same on the generic fibre by flat base change for global sections of a line bundle. We can therefore choose $\phi_i \in k(X)^\times$ with $\operatorname{div}_{X_\eta}(\phi_i|_{X_\eta}) = -r_i(K_{X_\eta} + \Gamma_i|_{X_\eta})$.

Write

$$K_X + \Gamma_i + \frac{1}{m_i} \operatorname{div}_X(\zeta_i) = f^* L_i, \qquad r_i \mid m_i,$$

with $\zeta_i \in k(X)^\times$. The ratio $\zeta_i / \phi_i^{m_i/r_i}$ has zero divisor on the generic fibre. It equals $f^*\theta_i$ for some $\theta_i \in k(Y)^\times$.[2] Put $H_{i,Y} = L_i - \frac{1}{m_i} \operatorname{div}_Y(\theta_i)$. Then

$$K_X + \Gamma_i + \frac{1}{r_i} \operatorname{div}_X(\phi_i) = f^* H_{i,Y}. \tag{3.2}$$

Here $H_{i,Y}$ is $\mathbb{Q}$-Cartier.

Take a log resolution $h_0 : W_0 \to X$ dominating the resolution in Step 2. Write Γ_{i,W_0} for the crepant boundary on W_0 and take a reduced snc divisor S_0 containing

$$\bigcup_i \operatorname{Supp} \Gamma_{i,W_0} \ \cup \ \bigcup_i \operatorname{Supp} \operatorname{div}_{W_0}(\phi_i) \ \cup \ \bigcup_i \operatorname{Supp} \operatorname{div}_{W_0}(\zeta_i).$$

Denote $g = f \circ h_0$. Remove from Y the images $g(D)$ of all vertical strata D of (W_0, S_0). For every horizontal stratum D, use generic smoothness to make $g|_D$ smooth over the remaining open subset. We also require this open subset to lie in the smooth locus of Y and g to be smooth over it. Furthermore, remove the supports of $C_{i,Y}$ and $H_{i,Y} - K_Y - C_{i,Y}$ for all i. Denote the resulting open subset by V_0. Here $C_{i,Y}$ is the discriminant trace for $f : (X, \Gamma_i) \to Y$, and its moduli trace $N_{i,Y}$ is defined using $H_{i,Y}$.

Resolve the ideal defining $Y \setminus V_0$ to obtain a smooth model $\psi : Y' \to Y$ such that $\psi|_V : V \cong V_0$, where $V = \psi^{-1}(V_0)$, and $\psi^{-1}(Y \setminus V_0)$ has snc support. Put

$$T = (\psi^{-1}(Y \setminus V_0))_{\mathrm{red}}, \qquad H_i = \psi^* H_{i,Y}.$$

The isomorphism $\psi|_V : V \cong V_0$ gives

$$C_{i,Y'}|_V = (\psi|_V)^*(C_{i,Y}|_{V_0}) = 0,$$

and

$$\begin{aligned} N_{i,Y'}|_V &= (H_i - K_{Y'} - C_{i,Y'})|_V \\ &= (\psi|_V)^*((H_{i,Y} - K_Y - C_{i,Y})|_{V_0}) \\ &= (\psi|_V)^*(N_{i,Y}|_{V_0}) = 0. \end{aligned}$$

[2]The components of the divisor of the ratio are vertical over Y. Remove the images of its poles to obtain a nonempty open subset of Y over which the ratio is regular on X. Since $f_* \mathcal{O}_X = \mathcal{O}_Y$, it is the pullback of a regular function on this open subset, which gives θ_i.

Thus the traces of $\mathbf{C}_i$ and $\mathbf{N}_i$ are supported on T for each i.

Finally, take the main component of $W_0 \times_Y Y'$ and resolve the inverse images of $\operatorname{Supp} T$ and $\operatorname{Supp} S_0$, leaving the inverse image of V unchanged. This gives a smooth model W with a morphism $f_W : W \to Y'$. The map $W \to W_0$ induces an isomorphism $f_W^{-1}(V) \cong g^{-1}(V_0)$. Put

$$S = (f_W^{-1}(\operatorname{Supp} T))_{\mathrm{red}} + \sum_{\substack{D \text{ a horizontal} \\ \text{component of } S_0}} \widetilde{D},$$

where $\widetilde{D}$ is the strict transform of D under $W \to W_0$. Pulling back (3.2) along $h : W \to X$ gives

$$K_W + \Gamma_{i,W} + \frac{1}{r_i} \operatorname{div}_W(\phi_i) = f_W^* H_i.$$

This is the desired diagram satisfying the four conditions of the resolved model.
Step 4.

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